Top 50 SQL Theoretical Interview Questions (Introduction Level)

SQL Basics

1. **What is SQL?**
   SQL (Structured Query Language) is used to store, retrieve, and manage data in relational databases.

2. **Why is SQL used?**
   SQL is used to efficiently access, manipulate, and analyze structured data stored in databases.

3. **What are the advantages of SQL?**
   SQL is easy to learn, fast, standardized, and capable of handling large datasets.

4. **Is SQL case-sensitive?**
   SQL keywords are not case-sensitive, but table and column names may depend on the database.

5. **Difference between SQL and NoSQL?**
   SQL uses structured relational data, while NoSQL handles unstructured or semi-structured data.

Database Concepts

6. **What is a database?**
   A database is an organized collection of structured data stored electronically.

7. **What is a table?**
   A table stores related data in rows and columns.

8. **What is a row?**
   A row represents a single record in a table.

9. **What is a column?**
   A column represents an attribute or field in a table.

10. **What is a relational database?**
    A database that stores data in related tables using keys.

Keys and Constraints

11. **What is a primary key?**
    A primary key uniquely identifies each record and cannot be NULL or duplicated.
12. **What is a foreign key?**
    A foreign key establishes a relationship between two tables.
13. **What is a composite key?**
    A key formed using two or more columns.
14. **What is a unique key?**
    Ensures unique values in a column but allows NULL values.
15. **What are constraints?**
    Rules applied to columns to ensure data integrity.

SQL Command Types

16. **What is DDL?**
    Data Definition Language (CREATE, ALTER, DROP).
17. **What is DML?**
    Data Manipulation Language (INSERT, UPDATE, DELETE).
18. **What is DQL?**
    Data Query Language (SELECT).
19. **What is DCL?**
    Data Control Language (GRANT, REVOKE).
20. **What is TCL?**
    Transaction Control Language (COMMIT, ROLLBACK).

Data Handling

21. **What is NULL?**
    Represents missing or unknown values.

22. **Difference between NULL and zero?**
    NULL means no value, zero is an actual value.
23. **What is DISTINCT?**
    Removes duplicate records from results.
24. **What is WHERE clause?**
    Filters records based on conditions.
25. **What is ORDER BY?**
    Sorts query results in ascending or descending order.

Aggregation and Functions

26. **What is GROUP BY?**
    Groups rows to apply aggregate functions.
27. **What is HAVING?**
    Filters grouped records after aggregation.
28. **What are aggregate functions?**
    COUNT, SUM, AVG, MIN, and MAX.
29. **What is COUNT()?**
    Returns the number of rows.
30. **What is LIKE operator?**
    Used for pattern matching in strings.

Transactions and Performance

31. **What is a transaction?**
    A group of SQL operations executed as a single unit.
32. **What is ACID property?**
    Atomicity, Consistency, Isolation, Durability.
33. **What is an index?**
    A database object that improves query speed.
34. **Why are indexes used?**
    To speed up data retrieval.
35. **What is a view?**
    A virtual table created using a SQL query.

Advanced Introduction Concepts

36. **What is a stored procedure?**
    A precompiled collection of SQL statements.
37. **What is a trigger?**
    Automatically executes when a database event occurs.
38. **What is data redundancy?**
    Unnecessary repetition of data.
39. **What is data consistency?**
    Ensures data remains accurate and uniform.
40. **What is normalization?**
    Organizing data to reduce redundancy.

SQL for Analytics and ML

41. **What is denormalization?**
    Combining tables to improve query performance.
42. **What is OLTP?**
    Systems designed for transactional processing.
43. **What is OLAP?**
    Systems designed for analytical processing.
44. **How is SQL used in data analysis?**
    For cleaning, aggregating, and exploring data.
45. **How is SQL used in ML?**
    To prepare features and training datasets.

Final Concepts

46. **What is schema?**
    The logical structure of a database.
47. **What is metadata?**
    Data about data.
48. **What is data integrity?**
    Accuracy and consistency of data.

49. **What is CHECK constraint?**
    Restricts values based on a condition.
50. **Can SQL handle large datasets?**
    Yes, SQL is designed to efficiently handle large datasets.